1 Notation and Symbols

1.1 Variables and Sets

t	Discrete time step index ($t = 0, 1, 2, \ldots$)
S_t	State at time t (random variable, uppercase)
s, s'	Particular state values (lowercase); s' denotes a next state
A_t	Action at time t (random variable)
a, a'	Particular action values
R_t	Reward received at time t (random variable)
r	A particular reward value
G_t	Return (cumulative discounted reward) from time t onward
T	Terminal time step (episodic tasks)
γ	Discount factor, $0 \leq \gamma \leq 1$
α	Step-size / learning rate parameter
θ	Convergence threshold for iterative algorithms
Δ	Maximum value change across a sweep (used in DP algorithms)
k	Iteration index in iterative algorithms
$\mathcal{S}$	Set of all states
$\mathcal{S}^+$	Set of all states including terminal state(s)
$\mathcal{A}$	Set of all actions
$\mathcal{A}(s)$	Set of actions available in state s
$\mathcal{R}$	Set of all possible reward values, $\mathcal{R} \subset \mathbb{R}$
$\lvert\mathcal{S}\rvert$	Number of states (cardinality of $\mathcal{S}$)
$\lvert\mathcal{A}\rvert$	Number of actions

1.2 Policies

π	A policy (mapping from states to action probabilities)
$\pi(a \mid s)$	Probability of taking action a in state s under policy π
$\pi(s)$	The action selected by a deterministic policy in state s
π_*	An optimal policy
π'	An improved policy (e.g., after greedy improvement)

1.3 Value Functions

$v_\pi(s)$	State-value function: expected return starting from s, following π
$q_\pi(s, a)$	Action-value function: expected return starting from s, taking a, then following π
$v_*(s)$	Optimal state-value function: $\max_\pi v_\pi(s)$
$q_*(s, a)$	Optimal action-value function: $\max_\pi q_\pi(s, a)$
$V(s)$	An estimate or array approximating a value function (mutable)
$V_k(s)$	Value estimate at iteration k

1.4 Probability and Dynamics Notation

$\mathbb{E}_\pi[\cdot]$	Expected value under policy π (agent follows π)
$\mathbb{E}_\pi[\cdot \mid S_t = s]$	Conditional expectation given state s at time t, following π
$p(s', r \mid s, a)$	**Dynamics function**: $\Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$
	Joint probability of next state s' and reward r, given current s and a
$p(s' \mid s, a)$	State-transition probability (dynamics marginalized over r)
$r(s, a)$	Expected reward for state-action pair
$r(s, a, s')$	Expected reward for state-action-next-state triple

1.5 Operators and Common Notation

$\sum_a$	Shorthand for $\sum_{a \in \mathcal{A}(s)}$
$\sum_{s',r}$	Shorthand for $\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}}$
$\max_a$	Maximize over all actions $a \in \mathcal{A}(s)$
$\arg\max_a$	The action a that achieves the maximum
$\leftarrow$	Assignment (update a stored value)
$\geq$ (on policies)	$\pi \geq \pi'$ means $v_\pi(s) \geq v_{\pi'}(s)$ for all s

2 Temporal-Difference (TD) Learning

Basic TD update rule:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[V(S_{t+1}) - V(S_t) \right] \tag{1}$$

where α is the learning rate (step size).

TD combines ideas from Monte Carlo (learning from experience) and Dynamic Programming (bootstrapping from estimates).

3 Finite Markov Decision Processes

3.1 Agent-Environment Interface

Agent and environment interact in discrete time steps: $t = 0, 1, 2, 3, \ldots$

Trajectory: $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots$

At each time t:

- Agent observes state $S_t \in \mathcal{S}$
- Selects action $A_t \in \mathcal{A}(s)$
- Receives reward $R_{t+1} \in \mathcal{R} \subset \mathbb{R}$
- Transitions to next state S_{t+1}

3.2 Dynamics Function

Complete probability distribution defining MDP dynamics:

$$p(s', r|s, a) = \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\} \tag{2}$$

for all $s', s \in \mathcal{S}$, $r \in \mathcal{R}$, $a \in \mathcal{A}(s)$.
Must satisfy: $\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r|s, a) = 1$ for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$.

3.3 Derived Quantities

State-transition probabilities:

$$p(s'|s, a) = \Pr\{S_t = s' \mid S_{t-1} = s, A_{t-1} = a\} = \sum_{r \in \mathcal{R}} p(s', r|s, a) \tag{3}$$

Expected reward for state-action pair:

$$r(s, a) = \mathbb{E}[R_t \mid S_{t-1} = s, A_{t-1} = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r|s, a) \tag{4}$$

Expected reward for state-action-next-state triple:

$$r(s, a, s') = \mathbb{E}[R_t \mid S_{t-1} = s, A_{t-1} = a, S_t = s'] = \sum_{r \in \mathcal{R}} r \frac{p(s', r|s, a)}{p(s'|s, a)} \tag{5}$$

3.4 Markov Property

Markov Property: The future depends only on the current state, not on the history.

$$\Pr\{S_t = s', R_t = r \mid S_0, A_0, R_1, \ldots, S_{t-1}, A_{t-1}\} = p(s', r|S_{t-1}, A_{t-1}) \tag{6}$$

A state signal has the Markov property if it includes all relevant information from history.

3.5 Returns

Episodic tasks (terminate in finite steps at terminal state):

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} + \cdots + R_T \tag{7}$$

where T is the terminal time step.
Continuing tasks (no terminal state, with discounting):

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots \tag{8}$$

where $0 \leq \gamma \leq 1$ is the discount rate.
Recursive form:

$$G_t = R_{t+1} + \gamma G_{t+1} \tag{9}$$

Special case (constant reward +1 forever):

$$G_t = \sum_{k=0}^{\infty} \gamma^k = \frac{1}{1 - \gamma} \tag{10}$$

3.6 Policies

Policy π: Mapping from states to probability distributions over actions.

$$\pi(a|s) = \Pr\{A_t = a \mid S_t = s\} \tag{11}$$

Deterministic policy: $\pi(s) = a$ (single action for each state).

3.7 Value Functions

State-value function for policy π:

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,\middle|\, S_t = s\right] \tag{12}$$

Action-value function (Q-function) for policy π:

$$q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \,\middle|\, S_t = s, A_t = a\right] \tag{13}$$

Relationship between value functions:

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a|s)\, q_\pi(s, a) \tag{14}$$

3.8 Bellman Equations

Bellman equation for v_π:

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_\pi(s')\right] \tag{15}$$

Bellman equation for q_π:

$$q_\pi(s, a) = \sum_{s',r} p(s', r|s, a) \left[r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s', a')\right] \tag{16}$$

3.9 Optimal Value Functions

Optimal state-value function:

$$v_*(s) = \max_\pi v_\pi(s) \quad \text{for all } s \in \mathcal{S} \tag{17}$$

Optimal action-value function:

$$q_*(s, a) = \max_\pi q_\pi(s, a) \quad \text{for all } s \in \mathcal{S}, a \in \mathcal{A}(s) \tag{18}$$

Relationship:

$$v_*(s) = \max_{a \in \mathcal{A}(s)} q_*(s, a) \tag{19}$$

$$q_*(s, a) = \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_*(s')\right] \tag{20}$$

3.10 Bellman Optimality Equations

Bellman optimality equation for v_*:

$$v_*(s) = \max_{a \in \mathcal{A}(s)} \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_*(s')\right] \tag{21}$$

Bellman optimality equation for q_*:

$$q_*(s, a) = \sum_{s',r} p(s', r|s, a) \left[r + \gamma \max_{a' \in \mathcal{A}(s')} q_*(s', a')\right] \tag{22}$$

3.11 Optimal Policy

Optimal policy (greedy with respect to q_*):

$$\pi_*(s) = \arg\max_{a \in \mathcal{A}(s)} q_*(s, a) \tag{23}$$

Alternative form (greedy with respect to v_* using one-step lookahead):

$$\pi_*(s) = \arg\max_a \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_*(s')\right] \tag{24}$$

Any policy that is greedy with respect to v_* or q_* is an optimal policy.

4 Dynamic Programming

DP methods require perfect model of environment: $p(s', r|s, a)$.

4.1 Policy Evaluation (Prediction)

Goal: Compute v_π for a given policy π.
Iterative update (until convergence):

$$V_{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a) \left[r + \gamma V_k(s')\right] \tag{25}$$

for all $s \in \mathcal{S}$.
Convergence criterion: Stop when $\max_{s \in \mathcal{S}} |V_{k+1}(s) - V_k(s)| < \theta$ for small threshold $\theta > 0$.
As $k \to \infty$, $V_k \to v_\pi$ under Bellman equation constraints.

4.2 Policy Improvement

Policy improvement theorem: Let π and π' be deterministic policies such that for all $s \in \mathcal{S}$:

$$q_\pi(s, \pi'(s)) \geq v_\pi(s) \tag{26}$$

Then $v_{\pi'}(s) \geq v_\pi(s)$ for all $s \in \mathcal{S}$.